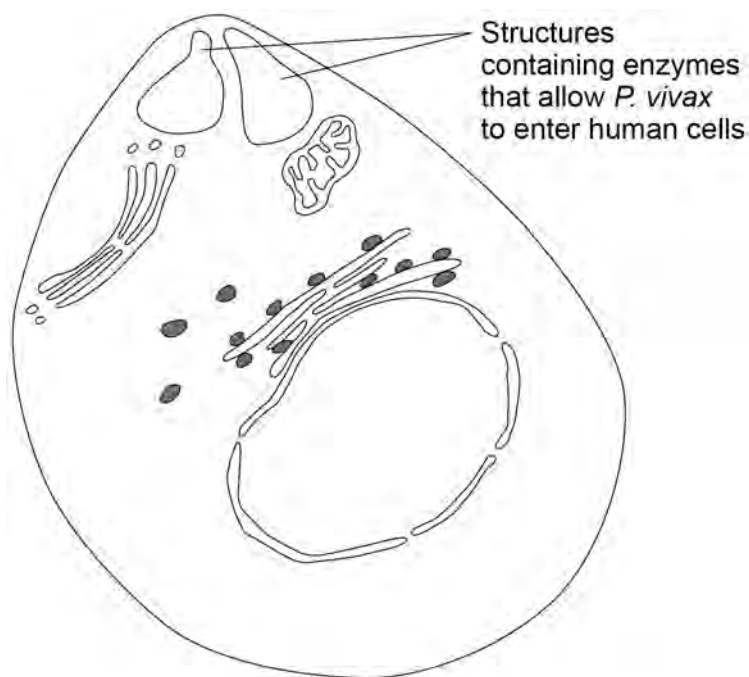


0 3 1

The human disease, malaria, is caused by infection with a single-celled eukaryotic organism.

Figure 2 shows a diagram of *Plasmodium vivax*, one of the species that can cause malaria.

Figure 2



Other than the Golgi apparatus, name **one** structure in **Figure 2** which shows that *P. vivax* is a eukaryote.

[1 mark]

0 3 2

Describe **two** functions of the Golgi apparatus in a eukaryotic cell.

[2 marks]

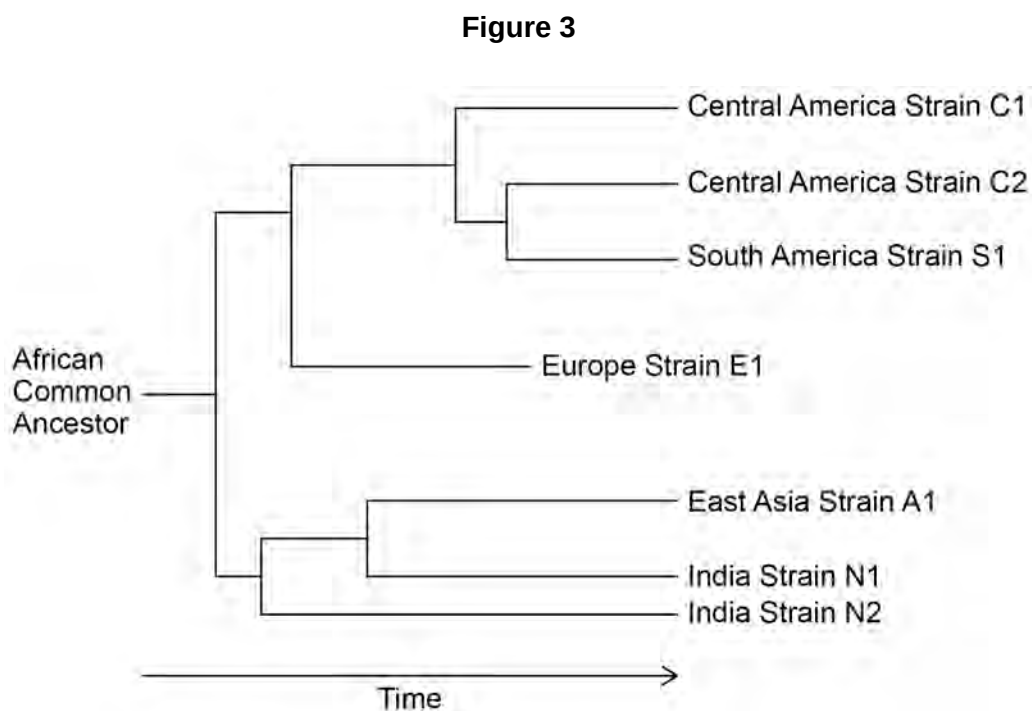
1

2

P. vivax evolved from a common ancestor in Africa. As humans migrated around the world, new strains of *P. vivax* evolved.

P. vivax is now extremely rare in Africa but there are several different strains of *P. vivax* in other parts of the world.

Figure 3 shows a phylogenetic diagram of the evolution of these different strains.



0 3 . 3 What does **Figure 3** suggest is the order of human migration out of Africa?

Tick (✓) **one** box.

[1 mark]

Europe, India, East Asia, Central America, South America

☐

India, East Asia, Europe, South America, Central America

☐

India, Europe, East Asia, Central America, South America

☐

South America, Central America, East Asia, Europe, India

☐

Question 3 continues on the next page

Turn over ►

[4 marks]

